Streamline Data Queries with LangChain

Understanding LangChain Queries

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Understanding LangChain

Open-source framework (Oct. 2022)

Develop custom AI applications on top of LLMs

LLMs + prompts + chains + Agents + tools

LCEL - LangChain Expression Language (compose chains)

Python + JavaScript

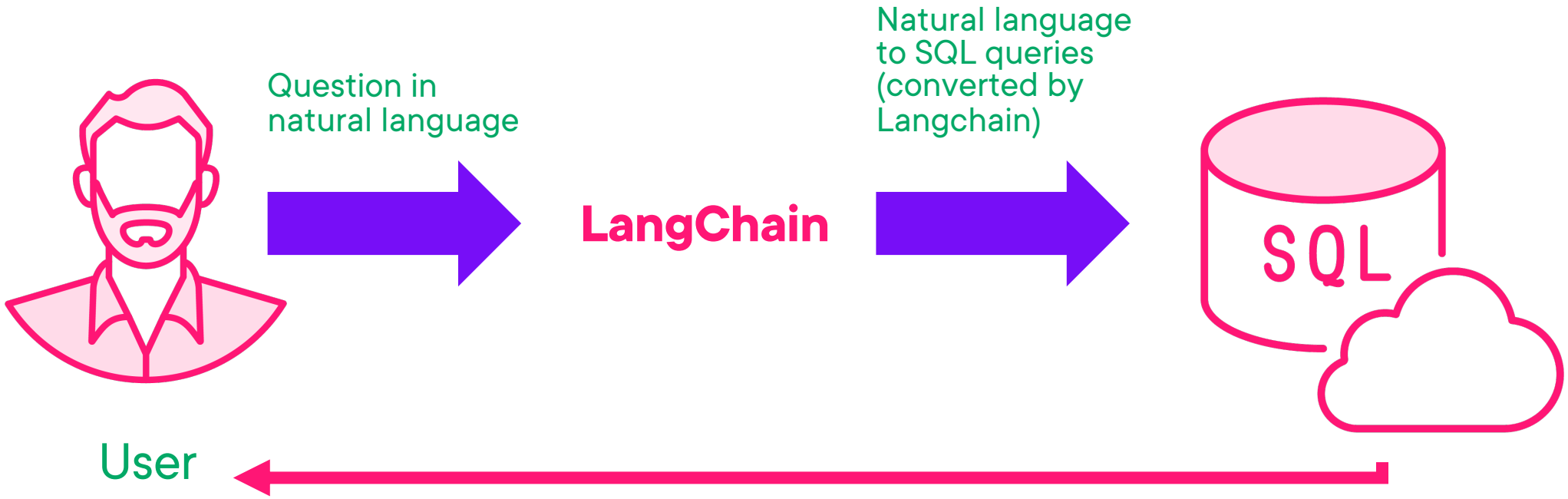
LangSmith (debugging, evaluating, monitoring)

Understanding LangChain Queries

LangChain Queries

LangChain queries are SQL queries generated automatically when a user asks questions in natural language and LangChain by leveraging LLMs to generate SQL queries automatically.

LangChain Queries Flow



LangChain Offers Two Ways to Build SQL Queries

SQL chains

SQL Agents

Three Types of Interaction Outcomes LangChain Allows with SQL Database

Text-to-SQL Query (Chains)

Text-to-SQL Query & Execution (Chains)

Text-to-SQL Query and Execution (Agents)

Advantages of LangChain Queries in Data Retrieval

Advantages of LangChain in Data Retrieval

Natural language: Business users can ask questions in natural language rather than SQL/SME dep.

Inexpensive: Getting data insights becomes inexpensive as large data team is not required

Easy to use & adopt: Since requests are made in natural language

Speed of insights: Removes data bureaucracy leading to speed of insights

Reduction in effort: LLMs automatically build queries, execute them, and generate insights

Data democratization: All business users can have access to data and insights

Demo

Enabling APIs

- OpenAI (for ChatGPT)
- LangSmith (for debugging)

Demo

LangChain queries: syntax & structure

- SQL Chain
- SQL Agents

Enabling Data Transparency Using LangChain Queries

Data Transparency

Explain-ability and interpretability

Clarity`

Ethics by detecting bias

Private data

Algorithms for Query Optimization

Remember About Algorithms with Regards to LangChain Queries

It should be noted that LangChain does not provide any built-in algorithms for LangChain queries for optimization.
It only provides techniques for optimizing LangChain queries.

LangChain Query Optimization

Popular Techniques of LangChain Query Optimization

- Caching
- Indexing
- Parallel & Distributed Execution
- Query Checker
- Custom Prompt

Demo

LangChain queries

- Indexing
- Caching

Demo

LangChain queries

- Parallel query execution
- Distributed query execution

Parallel and Distributed Query Optimization

LangChain Queries in Parallel

Though LangChain supports running “LLM chains” in parallel by RunnableParallel, but LangChain version 0.346 does not support parallel execution of multiple “SQL Chain” LangChain queries

```python
from langchain.agents import create_spark_sql_agent
from langchain.agents.agent_toolkits import SparkSQLToolkit
from langchain.chat_models import ChatOpenAI
from langchain.utilities.spark_sql import SparkSQL

from pyspark.sql import SparkSession


spark = SparkSession.builder.getOrCreate()
schema = “DB_LANGCHAIN"
spark.sql(f"CREATE DATABASE IF NOT EXISTS {schema}")
spark.sql(f"USE {schema}")
csv_file_path = “product_preferences.csv"
table = “table_product"
spark.read.csv(csv_file_path, header=True,
inferSchema=True).write.saveAsTable(table)
spark.table(table).show()
```

- Import Spark Agent
- Import Spark SQL Toolkit

- Initialize the Spark Session

- Create Schema
- Define File Path
- Define Spark table Name
- Load the File to Spark table

```
spark_sql = SparkSQL(schema=schema)
llm = ChatOpenAI(temperature=0)
toolkit = SparkSQLToolkit(db=spark_sql, llm=llm)
agent_executor = create_spark_sql_agent(llm=llm,
toolkit=toolkit, verbose=True)
```

```
Agent.executor(" How many customers from USA bought
products in Electronics category? ")
```

- Initialize LLM with temperature as 0
- Load the Spark SQL agent
- Run the Spark Agent Executor

Demo

LangChain queries

- Provide few shot examples
- Query checker